

Using Live Coding to Teach Econometrics with RISE

CEA2026 Conference

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- Switching between slides and a coding environment improves structure, but separates explanation from execution

A Better Approach: Live Coding with RISE

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- RISE is an extension that turns a Jupyter notebook into a slide deck
- One notebook, two uses:
 - Students receive a guided notebook with blanks to fill in
 - Instructors present the same notebook and code the solutions live

Demo: Introducing Loops to Economics Majors

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Suppose I want to plot a demand curve defined by:

$$P = 100 - 4Q$$

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Suppose I want to plot a demand curve defined by:

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For this, I need a list of quantities and corresponding prices.

Approach 1: Manual Approach

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```
In [1]: #not live coded
quantities = []
prices = []

#add quantity of 0 and corresponding price to a list
quantity = 0
price = 100 - 4 * quantity
quantities.append(quantity)
prices.append(price)

quantity = 2
price = 100 - 4 * quantity
quantities.append(quantity)
prices.append(price)

quantity = 5
price = 100 - 4 * quantity
quantities.append(quantity)
prices.append(price)
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Approach 2: Use a *loop* to calculate the price for each quantity

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```
In [3]: quantities = [0, 2, 5]
        prices = []

        #LIVE CODED
        for quant in quantities:
            price = 100 - 4 * quant
            prices.append(price)

        print(quantities)
        print(prices)
```

```
[0, 2, 5]
[100, 92, 80]
```

Adaptation to Teaching Econometrics

Jupyter Notebooks Only Natively Support Python/R/Julia

- For Stata support, there are two options: Stata API and nbstata

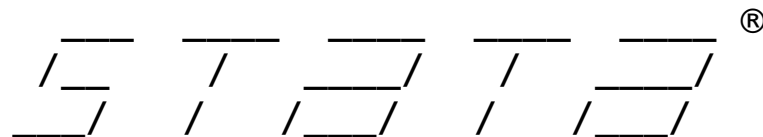
The Stata API is a Python package.

- Specify the location of Stata on your computer
- Then use %%stata at the beginning of cells and use Stata like normal

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```
In [4]: import stata_setup  
stata_setup.config("/Applications/Stata", "mp")
```



Statistics and Data Science

18.0

MP-Parallel Edition

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800-STATA-PC

070.606.4600

<https://www.stata.com>

stata@stata.com

In [5]: %%stata

```
sysuse auto, clear
regress price mpg
```

```
.
. sysuse auto, clear
(1978 automobile data)

. regress price mpg
```

Source		SS		df		MS		Number of obs	=	74	
-----+-----											
Model		139449474		1		139449474		F(1, 72)	=	20.26	
Residual		495615923		72		6883554.48		Prob > F	=	0.0000	
-----+-----											
Total		635065396		73		8699525.97		R-squared	=	0.2196	
									Adj R-squared	=	0.2087
									Root MSE	=	2623.7

price		Coefficient	Std. err.	t	P> t	[95% conf. interval]	
-----+-----							
mpg		-238.8943	53.07669	-4.50	0.000	-344.7008	-133.0879
_cons		11253.06	1170.813	9.61	0.000	8919.088	13587.03

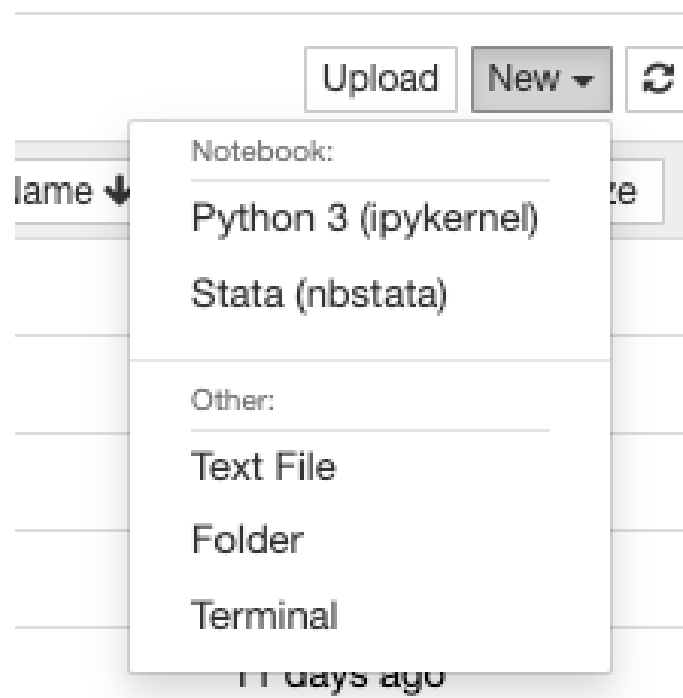
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In [1]: sysuse auto, clear
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In [2]: regress price mpg
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Resources



Scan for:

- Presentation notebook and PDF
- Sample live-coding lecture materials
- Anaconda/Jupyter setup resources
- RISE, `nbstata`, and Stata Python API documentation

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